

Question number	Answer	Mark
4(a)(i)	B myosin A is not correct because the structure labelled X is made from myosin not actin C is not correct because the structure labelled X is made from myosin not tropomyosin D is not correct because the structure labelled X is made from myosin not troponin	(1)

Question number	Answer	Mark
4(a)(ii)	D Z A is not correct because the structure labelled W does not change in length during contraction A is not correct because the structure labelled X does not change in length during contraction A is not correct because the structure labelled Y does not change in length during contraction	(1)

Question number	Answer	Mark
4(a)(iii)	D three A is not correct as all three statements are correct. B is not correct as all three statements are correct. C is not correct as all three statements are correct.	(1)

Question number	Answer	Mark
4(a)(iv)	B myosin A is not correct because actin does not have an enzyme activity that breaks down ATP C is not correct because tropomyosin does not have an enzyme activity that breaks down ATP D is not correct because troponin does not have an enzyme activity that breaks down ATP	(1)

Question number	Answer	Mark
4(b)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Indicative content Descriptive points - individual 1 no treatment stimulated normal oxygen consumption - individual 2 pyruvate all three treatments stimulated normal oxygen consumption - individual 3 succinate and reduced TMPD stimulated normal oxygen consumption - individual 4 only reduced TMPD stimulated normal oxygen consumption Linking treatments to activation of different points in etc - pyruvate and malate (produce reduced NAD) acts on (complex) I - succinate acts on (complex) II - reduced TMPD acts on cytochrome c Links to etc defect - idea that defects in different parts of etc - individual 1 defect is in or after cytochrome c - individual 2 defect is in before (complex) I / etc is intact - individual 3 defect is in (complex) I - individual 4 defect is in (complex) I, coQ or III (Ignore complex II)	(6)

Level	Marks	Descriptor
	0	No awardable content.
1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning with some structure.
3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.

